

Towards Multidisciplinary Citance Analysis: A Novel Approach for Intent, Polarity, and Semantics Classification

Georgios Pantelakis^{1,2} Petros Stavropoulos² Haris Papageorgiou²

¹Department of Informatics and Telecommunications,
National and Kapodistrian University of Athens, Greece

²Institute for Language and Speech Processing, Athena Research Center, Greece
gpantelakis@di.uoa.gr petros.stavropoulos@athenarc.gr haris@athenarc.gr

Abstract

The rapid increase in research publications, particularly in computer science, presents significant challenges for researchers, reviewers, and institutions in performing comprehensive literature reviews. Traditional citance analysis, which mainly focuses on sentiment and citance intent, often misses important details. To address this, we introduce a multidimensional citation-analysis ontology that jointly represents three complementary dimensions: Intent, Polarity, and Semantics. Using data from Semantic Scholar we classify citances based on this approach to create CitAn dataset which has 1,165 manually labeled citances from 80 subfields in Computer Science, Health Sciences, Bioinformatics, and Social Sciences and Humanities. Finally, we fine-tune language models using this dataset, which yields promising results and paves the way for efficient citance analysis across various academic fields.

1 Introduction

In today’s fast-growing academic landscape, particularly in fields such as computer science, an overwhelming number of research papers is published every day. For researchers starting new projects, conducting an effective literature review is essential for understanding prior work, identifying gaps, and assessing scientific relevance. Similar challenges affect peer reviewers, publishers, and academic institutions (A.15), which must evaluate an increasing volume of submissions for originality, rigor, and impact. These tasks require identifying contradictions, tracking extensions of foundational work, locating relevant datasets and tools, summarizing methods, and verifying novelty and reproducibility. Given the scale of modern scientific production, this process has become increasingly complex and time consuming.

Citance analysis, which studies the textual context in which references are used, provides a powerful solution to these challenges by revealing how

knowledge is used and developed within scholarly discourse. It has important implications for multiple stakeholders in the scholarly ecosystem. For researchers, it supports knowledge discovery, literature gap detection, and reproducibility. For research-performing organizations, it enables fine-grained impact assessment beyond raw citation counts. Publishers benefit from improved visibility into how articles are used and discussed, while funders gain clearer evidence of the scientific influence and effectiveness of their investments. These benefits highlight the growing importance of citance analysis in the modern research ecosystem. Traditionally, citance analysis has focused on two main aspects: sentiment and citation function. While useful, these isolated perspectives often fail to capture the full context of how and why scientific works are cited. In this work, we address this limitation by integrating intent, polarity, and semantics into a unified analysis framework. Our approach leverages citance data from Semantic Scholar and the semantic capabilities of language models.

We introduce a novel ontology that classifies citances based on three complementary dimensions: Intent, Polarity, and Semantics. This framework enables a deeper understanding of the author’s purpose in citing a work, the author’s stance toward it and the object of the reference. In addition, we introduce a dataset of 1,165 manually annotated citances from three major scientific domains: Computer Science, Health Sciences and Bioinformatics, and Social Sciences and Humanities, covering 80 subfields as detailed in Appendix A.9. This dataset is used to fine-tune language models for domain-agnostic citance analysis. To the best of our knowledge, no prior work jointly models citation intent, polarity, and semantics at the level of an individual target citation marker, nor is there a publicly available, domain-agnostic dataset that captures all three dimensions. The task addressed in this paper is illustrated in Figure 1. Our main contributions are

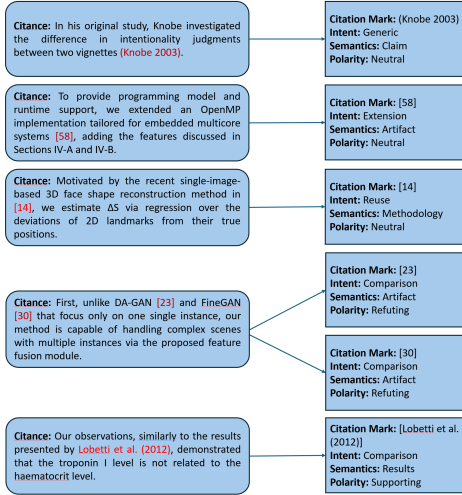


Figure 1: Example citances and their corresponding classifications across the Semantics, Intent, and Polarity categories, based on the ontology defined in this work. Each citance mark in the same sentence gets its own annotation.

as follows: (i) we propose a multidimensional ontology for citance analysis based on Intent, Polarity and Semantics; (ii) we introduce the CitAn dataset, containing 1,165 manually annotated citances spanning 80 subfields and three broad scientific domains; (iii) we provide benchmark experiments with several language-model architectures demonstrating the usability of the proposed resource for automated citance analysis; (iv) we present a qualitative evaluation of the best-performing model. The ontology, annotated dataset, and associated resources are available through our GitHub repository.¹

2 Related Work

Currently, research in the area of citance analysis has primarily focused on two branches: the development of ontologies for marking references and citance data, and the development of automated tools for annotation based on these ontologies. A brief review of the works that inspired us and are most closely related to our research is provided. For a more detailed review of the literature over the years on ontology definition, refer to (Ihsan and Qadir, 2019) and for automated classification of reasons of the citation to (Ihsan et al., 2023).

¹<https://github.com/pantelakis/multidisciplinary-citance-analysis>

2.1 Ontology

Prior research on citance analysis has proposed a variety of ontologies and classification schemes, yet many remain limited in how they conceptualize relationships between citing and cited works (Ciccarese et al., 2008; Zhu et al., 2015; Valenzuela-Escárcega et al., 2015). Others use large or hierarchical label inventories (Peroni and Shotton, 2012; Zhao et al., 2019), or combine and overlap categories that could be represented more simply as independent dimensions (Ihsan and Qadir, 2019; Budi and Yaniasih, 2023). These limitations motivate the need for a simpler but comprehensive framework for citance analysis.

The SWAN ontology (Ciccarese et al., 2008) categorizes research through discourse elements like Research Statements and Structured Comments, linked by tags such as "discusses" or "inconsistentWith." Unlike our approach, it emphasizes internal connections rather than relationships between a paper and its citances.

CiTO (Peroni and Shotton, 2012) represents citation relations through 32 citation-relation types, including properties such as *usesMethodIn*, *usesDataFrom*, *extends*, *agreesWith*, and *disagreesWith*. These properties overlap with CitAn but operate at a different level: for example, *usesMethodIn* combines Reuse Intent with Methodology Semantics, whereas CitAn annotates these dimensions separately and independently adds Polarity. This factorization avoids requiring a separate relation property for every combination of purpose, cited object, and stance.

Works like (Zhu et al., 2015), (Valenzuela-Escárcega et al., 2015) classify citances by importance using a two-level system, focusing on background or extended contributions. However, they overlook citance intent and underlying author motivation, which are often subjective and domain-specific.

CCRO (Ihsan and Qadir, 2019) surveyed approximately 150 citation reasons and reduced them to eight reason properties grouped by sentiment: BasedOn, Incorporate, and Extend under Positive; Negate, Criticize, and Contrast under Negative; and Compare and Discuss under Neutral. Its reason properties therefore combine information that CitAn represents separately as Intent and Polarity, and CCRO does not annotate the cited object captured by CitAn’s Semantics dimension.

(Zhao et al., 2019) introduced SciRES, which

has 18 label values: three general roles, nine fine-grained roles, and six functions. Its general roles are Material, Method, and Supplement; Supplement contains the fine-grained roles Paper, License, and Media, so it does not map directly to CitAn’s Claim label. Use and Produce are two separate function labels: Use concerns applying an existing resource, whereas Produce concerns producing or releasing a resource. Use, Extend, and Compare correspond closely to CitAn’s Reuse, Extension, and Comparison Intent labels, while Produce can be represented through CitAn’s Generic Intent and Artifact Semantics. Unlike CitAn, SciRES has no Polarity labels and is restricted to online-resource citations.

In 2023, (Budi and Yaniasih, 2023) introduced a citance classification scheme with 12 label values: three sentiment, four role, and five function labels. This is the closest prior scheme to CitAn’s three-dimensional formulation: sentiment corresponds to Polarity, role to Semantics, and function to Intent. However, the individual labels do not align one-to-one. For example, their Utilizing and Comparing functions correspond closely to Reuse and Comparison, whereas their Introducing, Relating, and Explaining functions are all covered by CitAn’s Generic label; conversely, CitAn includes an explicit Extension label.

(Hariri, 2024) analyzed sentiment in academic writing across disciplines, inspiring our polarity definition by highlighting domain-specific citance expressions.

Table 1 summarizes the specific overlaps and differences between CitAn’s and other works.

2.2 Data

The most frequently utilized corpus in the field is ACL Anthology², which features a collection of more than 100,000 papers dedicated to research in computational linguistics and natural language processing. (Athar, 2011) manually annotated a corpus of 8,736 citances from 310 papers extracted from ACL ARC (Bird et al., 2008) dataset, which is a manually curated and standardized subset of ACL Anthology, using positive, negative, and objective labels. Extending their work, (Athar and Teufel, 2012) created a new corpus consisting of 1,741 citances by utilizing a four-sentence context window. Since the long context window could yield multiple sentiments per citance, the authors selected the

sentiment found in the latest sentence, as previous work suggests this is most likely to reflect the author’s intent (MacRoberts and MacRoberts, 1984). Additionally, they introduced an "excluded" label to exclude sentences without any direct or indirect mention of the citance.

(Valenzuela-Escárcega et al., 2015) annotated 465 citances from the ACL Anthology with four intent classes, based on an ontology of incidental and important citations. SciCite (Cohan et al., 2019), a larger, multi-domain corpus annotated for citance intent, lacks a sentiment perspective. (Zhao et al., 2019) introduced the SciRes dataset with 3,088 annotated resource contexts, categorizing citances by role and function, though it is not multidisciplinary.

Our dataset significantly advances the current state of research by integrating annotations for intent, polarity, and semantics into each citance, a unique approach, as this combination has not been explicitly addressed in existing datasets to our knowledge. We ensure a balanced representation by frequently covering almost all combinations of these three categories. Moreover, our dataset spans a broad domain, encompassing three major fields and 80 subfields, providing comprehensive coverage. With 1,165 data points that combine intent, polarity, and semantics, our dataset offers a rich, detailed view of the full spectrum of information available in these areas.

2.3 Evaluation Challenges

Although several datasets could potentially be used to evaluate our work, such as the polarity-focused dataset by (Athar, 2011), the SciCite dataset from (Cohan et al., 2019), and the SciRES dataset from (Zhao et al., 2019), they differ significantly in their underlying ontologies and are thus semantically misaligned, making direct comparisons infeasible. Even when label names appear similar to those used in this work, their meanings often diverge, rendering comparisons misleading. Additionally, some of these datasets include sentences with multiple citances without clarifying which citance the label refers to or whether it applies to all, leading to ambiguous interpretations. In contrast, our labels are consistently assigned with respect to both the citing author’s perspective and a specific citation marker, a detail not made explicit in the other datasets. Our work also combines three distinct annotation categories within a unified framework, an approach not explored in prior benchmarks. Since our models are trained to reason jointly across all

²<https://aclanthology.org/>

Work	Total values	Closest overlap with CitAn	Difference from CitAn
Peroni and Shotton (2012) (CiTO)	32 relation types	<i>usesMethodIn/usesDataFrom</i> → Reuse + Methodology/Artifact <i>extends</i> → Extension <i>agreesWith/supports; disagreesWith/refutes</i> → Supporting; Refuting	CiTO enumerates a larger set of fine-grained, partly overlapping citation relations (disagreesWith; disputes; refutes; critiques), whereas CitAn represents similar distinctions compositionally using 11 factorized labels.
Ihsan and Qadir (2019) (CCRO)	8 reason labels	<i>Incorporate; BasedOn</i> → Reuse <i>Compare</i> → Comparison + Neutral <i>Contrast</i> → Comparison + Refuting <i>Negate; Criticize</i> → Refuting	- CCRO groups its reasons under Positive, Negative, and Neutral branches, coupling function with sentiment and splitting similar relations across branches. - CitAn factorizes Intent and Polarity into reusable labels and adds a separate Semantics dimension.
Cohan et al. (2019) (SciCite)	3 labels	<i>Background</i> → Generic <i>ResultComparison</i> → Results or Comparison <i>Method</i> → Methodology or Artifact	Its three-class scheme is simple but covers only Intent and overlaps categories; CitAn remains compact while also representing Semantics and Polarity.
Zhao et al. (2019) (SciRES)	18: 3 general roles + 9 fine roles + 6 functions	<i>Data/Tool/Code</i> → Artifact; <i>Algorithm</i> → Methodology <i>Use</i> → Reuse; <i>Extend</i> → Extension <i>Compare</i> → Comparison; <i>Introduce</i> → Generic (partial)	- SciRES separates resource roles from citation functions, but focuses on resource types and does not cover CitAn’s broader Semantics categories, particularly Claim and Results. - It has no Polarity dimension and is restricted to online-resource citations.
Budi and Yaniasih (2023)	12: 3 sentiment + 4 role + 5 function	<i>Positive/Neutral/Negative</i> → Supporting/Neutral/Refuting <i>Utilizing</i> → Reuse; <i>Comparing</i> → Comparison <i>Data/Method/Result</i> → Artifact/Methodology/Results	Its structure closely parallels CitAn, but its categories are coarser: it lacks an Extension function, uses broad introductory and relational functions, and includes a generic Supplemental role rather than explicitly distinguishing Claim as a cited semantic element.

Table 1: Overlap and differences between CitAn and related schemes. CitAn is multidisciplinary and has 11 values (4 Semantics, 4 Intent, and 3 Polarity). CitAn labels a target citation marker rather than a sentence or broader context.

three categories, evaluating them on datasets that address only a single aspect would require isolating components of the system. This is not only methodologically unfair but also undermines the integrated nature of our approach. We proposed a tentative label mapping in Appendix A.12, but due to the issues outlined above, it cannot currently be considered meaningful or reliable. We also attempted to contact the authors of the referenced datasets for clarification but did not receive any responses. Given these fundamental differences in ontology, structure, and perspective, we conclude that a fair comparison is not currently feasible.

3 Methodology

In this section, we describe our approach to analyzing and categorizing citances by describing the ontology used in this work, where we focused on three key aspects: Intent, Polarity, and Semantics. To create our dataset, we extracted citances from a variety of research papers that we filtered using various techniques. Subsequently, the author manually curated the dataset to ensure the validity of it. Finally, we describe the fine-tuning process of various language models for our experiments.

3.1 CitAn Ontology

Current work on the ontology of citance analysis either focused on polarity or the intent or when combined had many different categories. Studying the relative work and through experimentation, our findings indicated that most proposed categories were subsumed within the existing ones and did not provide substantial additional value. While considering potential additional categories from the ones we currently include in our method, we conducted extensive experiments to evaluate their viability. Additionally, while our goal was to provide an easy to comprehend annotation ontology and versatile as possible, we also wanted to create categories that can have a meaning both individually and collectively.

The three dimensions are designed to capture complementary aspects of a citation. Intent answers why the author cites a work, Semantics identifies what aspect of the cited work is being referenced, and Polarity describes how the citing author positions themselves toward that work. Keeping these dimensions separate avoids conflating citation function, referenced object, and evaluative stance, while their joint annotation provides a richer description of citation usage. For example, two citances may both reference a methodology but differ because one reuses it neutrally whereas

another critiques it. Such distinctions cannot be fully captured when any one of citation intent, sentiment, or semantic role is considered in isolation. The practical value of preserving this granularity is also supported by a stakeholder evaluation within the EU-funded TIER2 project, where stakeholders from research funding and research-performing organisations emphasised the importance of maintaining granularity when combining indicators (Leitner et al., 2025).

Intent encompasses the author’s purpose behind referencing the cited paper. Four distinct categories were defined under Intent: (i) **Reuse**: The author directly uses elements from the cited work in their research. (ii) **Comparison**: The author compares their work directly with the cited work, highlighting similarities or differences. (iii) **Extension**: The author’s research builds upon and enhances the cited work to advance their study. (iv) **Generic**: The author provides general information with or without direct relation to their own work.

Polarity reflects the author’s sentiment towards the cited paper, categorized into three classifications: (i) **Supporting**: The author agrees with and positively references the cited work. This can be expressed through technical acknowledgment or positive appraisal. Indicators include agreeing with findings, endorsing conclusions, or highlighting the benefits or strengths of the cited work. (ii) **Neutral**: Author cites the work without expressing a stance. (iii) **Refuting**: Author disagrees with or challenges the cited work. The refuting sentiment could also be expressed through the technical disregard of the cited work.

Semantics relates to the main topic or object referenced within the citance. Four categories were established: (i) **Claim**: Citances refer to specific statements, theories, main points or conclusions from the cited paper. They are used to back up, question, or provide context for what the author is saying. (ii) **Results**: Citances refer to the actual findings, data analysis, or results from the cited paper. These references point out specific evidence or discoveries of the cited paper. (iii) **Methodology**: Citances refer to the procedures, techniques, experimental designs, or analytical methods employed in the cited research. These are used to explain, adopt or critique the methodological approaches of the cited study. (iv) **Artifact**: Citances refer to tangible or intangible resources, tools, blueprints, models, datasets, software, lab equipment, surveys, medicine or other assets developed or utilized in

the cited work. These references highlight specific materials or instruments that are essential for conducting research, implementing solutions, or enabling further studies.

3.2 CitAn Dataset

The dataset introduced in this work is multidisciplinary and built on a structured ontology that captures rich information from citances, offering a strong foundation for future research. To reduce bias, we included papers beyond just highly influential ones, recognizing that influence and a positive view may correlate with author experience. This approach enables deeper analysis and opens new research directions. The following subsections detail the dataset creation process.

Design Principles: CitAn was constructed according to four design principles: (i) **multidimensionality**, by separately annotating what is cited (Semantics), why it is cited (Intent), and the citing author’s stance toward it (Polarity); (ii) **marker-level annotation**, by assigning labels to each target citation marker rather than to the sentence as a whole; (iii) **categorical distinctness**, by using a limited set of operationally distinct categories that remain meaningful both independently and jointly; (iv) **multidisciplinary and diverse coverage**, by sampling across three broad domains, 80 subfields, and quantile-based combinations of publication year, citation count, and influential-citation count.

Extraction: The extraction process began by randomly selecting 5,000 papers from three distinct domains: Computer Science, Health Sciences and Bioinformatics, and Social Sciences and Humanities. This selection was made through Semantic Scholar’s domain filters.

Sampling: Initially, we calculated the distributions of year, citation count, and influential citation count³. Each feature was divided into five quantile-based bins (5^3 combinations), though only 45 were actually present due to sparsity or overlaps. From these, we sampled to ensure at least one record per combination. Papers with zero citations or references were excluded. The final dataset contained 4,040 papers and 200,000 citance and reference sentences.

Llama 3.1 filtering & Manual Annotation: To improve the efficiency and accuracy of identifying relevant citances among the 200,000 data points for

³Semantic Scholar API: citations where the cited work had significant influence on the citing publication.

manual curation, we employed a quantized Llama 3.1-8B model (Grattafiori et al., 2024), which was configured with a system prompt aligned with our ontology (details presented in the Appendix A.1), aiding in the classification of each data point’s semantics, intent, and polarity. Following this automated classification, the first author manually curated citances of interest and adjusted their assigned category (automation bias analysis available in A.20). During this process, we aimed to maintain a balance across both fields of study and the ontology categories. This involved iteratively sampling and verifying examples from underrepresented categories, removing inconsistent or low-quality entries, and refining the dataset until a reasonably balanced and representative set of 1,165 citances was obtained. Subsequently, by looking at the distribution we created extra synthetic data using ChatGPT (GPT-4o), to balance the less frequent categories or some that had no data at all (prompt and class combinations created at Appendices A.3, A.10).

Dataset Statistics: The CitAn dataset contains 1,165 citances from 80 unique fields of study, covering 43 of 48 possible classification category combinations, with missing cases mainly involving supporting and refuting polarities. It includes 430 Computer Science (CS) citances (36.91%), 456 Health Sciences and Bioinformatics (HSBI) citances (39.14%), and 279 Social Sciences and Humanities (SSH) citances (23.95%). Descriptive statistics are shown in Table 2. CS and HSBI exhibit similar polarity and intent distributions, although HSBI shows slightly higher Reuse (23.03%). In contrast, SSH has more Neutral cases (77.78%), higher Generic intent (60.57%), and lower Comparison, Reuse, and Extension frequencies. In the Semantics category, Artifact and Methodology dominate CS (81.63%), HSBI shows a more balanced distribution (56.36%), and SSH has the highest Claim frequency (40.86%), with Artifact and Methodology accounting for only 35.48% (see Appendix A.11).

Lower Methodology and Artifact rates in SSH may reflect differing research practices. Unlike CS and HSBI, SSH often relies on qualitative methods such as interviews and surveys, which involve fewer reusable datasets, tools, or software. This may explain the lower Reuse and Extension frequencies and reduced Artifact presence. The higher proportion of Neutral citances may also reflect disciplinary norms favoring neutral citation tone and indirect expression of criticism or praise.

3.3 Inter-annotator agreement

To assess the reliability of the annotation process, we computed inter-annotator agreement across the three annotated dimensions: Semantics, Intent, and Polarity. Agreement was measured between annotator A, who reviewed the initial Llama 3.1 output and adjusted labels where needed, and annotator B, who is an ongoing MSc student in Artificial Intelligence, had not seen the Llama 3.1 outputs, using only the annotation guidelines. Although multi-labeling was allowed, no such cases were observed (see Appendix A.17 for further discussion). Agreement was measured using both raw percentage agreement and Cohen’s Kappa (κ), which adjusts for chance agreement. For Semantics, the annotators achieved moderate agreement (58.54%, $\kappa = 0.4386$). The main source of disagreement involved Artifact and Methodology labels, with 173 cases where annotator A’s label was Artifact but B selected Methodology. Occasional confusions were also observed between Claim and Methodology (120 cases). For Intent, agreement was substantially higher at 84.3% ($\kappa = 0.76$). Annotators showed strong consistency in identifying Generic Intent and Comparison, with most disagreements involving Generic being confused with Reuse or Comparison. For Polarity, agreement reached 81.2% ($\kappa = 0.61$). Most disagreements arose between Neutral and Supporting categories (87 instances where A’s label was Neutral but B’s Supporting), while Refuting labels showed the highest consistency (141 agreements). The relative difficulty in distinguishing Neutral from weakly Supporting statements likely contributed to the lower κ compared to Intent.

Based on these findings, and following adjudication by a third annotator (a PhD candidate in AI with a focus on NLP), ambiguous cases were reviewed and resolved. This process led to the creation of an updated dataset, reflecting these adjudicated corrections. Overall, the results indicate substantial reliability for Intent and Polarity, while Semantics remains more challenging due to the difficulty of interpreting sentences without surrounding context. For detailed analyses and confusion matrices, refer to Appendix A.16.

3.4 CitAn Fine-Tuned Models

For the task of citance analysis, we experimented with various language models. Following prior work (Athar, 2011), we replaced citance markers

Intent	Count	Percentage	Semantics	Count	Percentage	Polarity	Count	Percentage	Synthetic	Count	Percentage
Comparison	225	19.31	Artifact	346	29.70	Neutral	826	70.90	Non-Synthetic	1,040	89.27
Extension	100	8.58	Claim	259	22.23	Refuting	156	13.39	Synthetic	125	10.73
Generic	603	51.76	Methodology	361	30.99	Supporting	183	15.71			
Reuse	237	20.35	Results	199	17.08						
Total	1,165	100	Total	1,165	100	Total	1,165	100	Total	1,165	100

Table 2: Descriptive statistics of the CitAn dataset for the categories: Intent, Semantics, Polarity, and Synthetic distribution. Numbers under percentage columns are in %.

in both the training and inference datasets with the token <CIT>. This step was taken to eliminate potential bias from researcher names and direct the model’s attention to the <CIT> token during training and inference. For the SciBERT architecture, we replaced citance marks with the [MASK] special token. All models were trained on the training split (705 citances), the only partition containing synthetic data (125 citances). Performance was monitored on a separate evaluation split (216 citances), and final results were obtained on the test set (244 citances). Detailed dataset splitting procedures are provided in A.6 and A.7, learning curve analysis A.19, impact of synthetic data A.21 and details on training hyperparameters and configurations are provided in Appendix A.8.

The first is Llama 3.1 (Grattafiori et al., 2024). In selecting Llama 3.1 for fine-tuning, we prioritized access to a high-performance model that combines the benefits of open-source availability with competitive benchmarking capabilities. We utilized a quantized version of Llama 3.1 to reduce memory usage and for faster computations. It was coupled with QLoRA (Quantized Low-Rank Adaptation) (Dettmers et al., 2023), which enabled us to perform fine-tuning more efficiently by applying quantization techniques to a select subset of model parameters. We also used Unsloth⁴ library, which provides a streamlined and efficient framework for training language models on specialized datasets with minimal computing resources.

Additionally, we finetuned a Flan-T5 model (Chung et al., 2024), which is an encoder-decoder fine-tuned model from the T5 family. Before fine-tuning, the data were formatted to input, output pairs as following: input: "Question: .. Context: ..", output: "Semantics – Intent – Polarity". For the input, Question included a short instruction prompt (prompt presented in the Appendix A.2), Context included the citance sentence, while output included the label of each category for that sentence.

As a final approach, we designed a custom multi-

task architecture based on SciBERT model (Beltagy et al., 2019), which extends BERT (Devlin et al., 2019) through pre-training on large-scale scientific text from Semantic Scholar. As illustrated in Figure 2, the embedding of the [MASK] token, used to mark the target citance, is passed to a shared multilayer perceptron (MLP) for feature compression. The resulting representation is then fed into three task-specific MLP heads for semantics, intent, and polarity classification. ReLU activations are applied to introduce non-linearity. The model is trained in a multi-task setting using a custom loss that minimizes the combined loss across all tasks. Performance is evaluated at each epoch by maximizing the overall F1 score, computed as the sum of the task-specific F1 scores.

4 Results

The newly developed ontology addresses a key gap in the literature by integrating three distinct categories into a comprehensive classification scheme. The manual labeling of the dataset using the proposed ontology was efficient and straightforward for annotators, and the categories were sufficiently distinct to minimize overlap, enabling consistent annotation and reliable downstream modeling. Furthermore, the ontology demonstrated broad applicability across various fields of study. Initially, we experimented with few-shot LLMs A.4. Subsequently, we fine-tuned Flan-T5, Llama 3.1, and a custom SciBERT multitask classification architecture using our dataset.

The resulting performance on Semantics, Intent, and Polarity is presented in Table 3. As the few-shot approaches yielded inferior results compared to fine-tuning, their performance is reported in Appendix A.5. To provide a comparison with our fine-tuned models, we also evaluated the base versions of Llama 3.1 and GPT-4o mini. The Flan-T5 Base model, without fine-tuning, is unable to generate the required output and is therefore not considered in the comparisons. For these evaluations, we used a system prompt that describes our ontology (see Appendix A.1).

⁴<https://github.com/unslothai/unsloth>

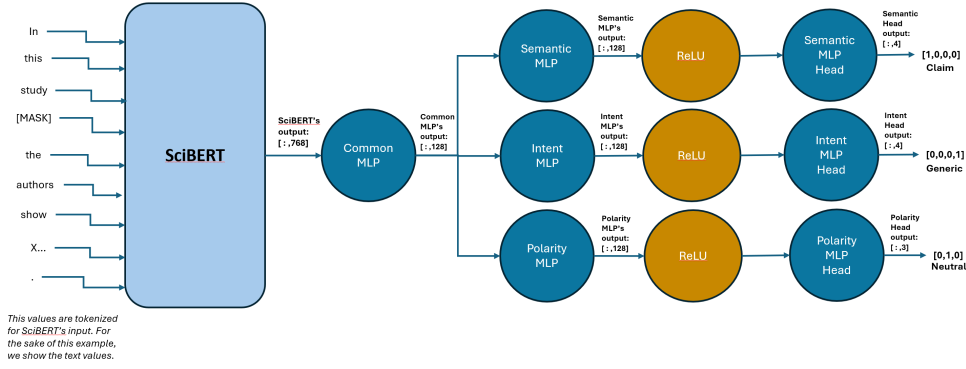


Figure 2: Architecture of Custom SciBERT Multi-Task classifier.

In semantic classification, Llama 3.1-FT achieved the highest scores across all metrics. The relatively small gap between macro and weighted F1 suggests that the model performs well across both frequent and rare classes. SciBERT-MT followed closely, demonstrating strong overall performance. For intent classification, SciBERT-MT outperformed all other models, achieving the highest macro and weighted F1 and accuracy. Its domain-specific pretraining likely contributes to its strong ability to capture rhetorical patterns and citation intent. For polarity classification, Llama 3.1-FT achieved the highest accuracy, macro F1, and weighted F1, indicating the strongest overall performance across both frequent and rare polarity classes.

A comparison with the non-fine-tuned models demonstrates the substantial impact of task-specific fine-tuning. For example, Llama 3.1 improves from 41.70% to 80.90% weighted F1 on the semantic task after fine-tuning, with similar gains observed for intent and polarity. Additionally, we compared the base and fine-tuned LLaMA models using McNemar’s test (McNemar, 1947) on paired predictions (A.18). The results show that fine-tuning leads to a statistically significant improvement across all three categories ($p < 0.001$). These results confirm that the performance improvements are driven by effective adaptation to the CitAn dataset and the proposed ontology. More detailed analyses by field and by class are provided in Sections A.13 and A.14, showing that Llama 3.1-FT is strongest in the Social Sciences and Humanities, while results are task-dependent in Computer Science and Health Sciences and Bioinformatics.

5 Qualitative Analysis

In this section, we evaluate the performance of the fine-tuned Llama 3.1 model across the three classification dimensions: semantic, polarity, and intent. Through representative prediction examples, we highlight both the model’s strengths and its limitations. In the following examples, bold text indicates agreement with the gold label, and an asterisk (*) marks a predicted label that differs.

As shown in Figure 3, the model struggles with citation-specific interpretation in complex sentences. Although both ResNet and ResNeXt are artifacts, the model incorrectly classifies them as Methodology, likely due to the procedural cue "by branching". For intent classification, the presence of a comparative structure causes the model to predict Comparison rather than Generic, even though the authors are not directly comparing their own work to the cited methods. In addition, polarity is predicted as Refuting for both citations, despite only ResNet being framed negatively. This example highlights the model’s difficulty in separating roles and sentiment when multiple references appear within the same comparative context.

In contrast, Figure 4 demonstrates a successful prediction. Although the sentence begins with positive language, it ultimately emphasizes the limitations of the artifact, resulting in an overall Refuting polarity. The model correctly captures this shift, showing its ability to detect implicit critique despite initial praise. Similarly, in Figure 5, the model correctly classifies both citations as Methodology, even in the presence of strong extension-related cues. This indicates that the model can recognize when an extension is attributed to prior work rather

Model	Semantic			Intent			Polarity		
	macro F1	weighted F1	accuracy	macro F1	weighted F1	accuracy	macro F1	weighted F1	accuracy
Llama 3.1-FT	80.66	80.90	81.15	74.55	77.73	77.87	63.89	72.92	77.46
SciBERT MT Class	77.39	78.25	78.28	79.41	81.42	81.97	58.59	69.04	75.41
Flan T5 Base FT	66.28	64.48	64.75	67.81	74.85	74.18	58.97	69.95	74.18
GPT-4o mini	58.34	58.94	59.43	64.55	66.73	67.21	58.71	62.58	60.65
Llama 3.1 Base	41.56	41.70	52.46	59.75	57.02	58.61	50.42	63.99	64.34

Table 3: F1 scores (macro and weighted) and accuracy for each language model across semantic, intent, and polarity tasks on the CitAn dataset. Numbers are reported in % and calculated on the test set. Statistical analysis of metrics is provided in Appendix A.22.

than to the authors themselves.

Finally, Figure 6 highlights the model’s ability to identify subtle lexical cues such as "adapted from", correctly classifying the citation as Reuse rather than Generic. The model also recognizes that the cited content describes a procedural step, leading to an accurate Methodology label rather than Results or Artifact. Additionally, it avoids hallucinating sentiment and correctly assigns a neutral polarity.

Overall, the model shows a strong ability to interpret academic writing and capture meaning beyond surface-level cues. However, it still struggles with sentences that contain multiple references, where overlapping signals make interpretation more difficult.

6 Conclusions and Future work

In this work, we propose an ontology for analyzing citances based on intent, polarity, and semantic categories, along with the manually annotated CitAn dataset built on this framework. Using this data, we fine-tuned language models for automated citance classification and achieved promising results. While SciBERT achieved the best performance on intent classification, the fine-tuned Llama 3.1 model achieved the highest performance on semantic and polarity tasks among all evaluated models. Statistical significance testing using McNemar’s exact test confirms that these gains are significant with respect to the base Llama 3.1 model across all three categories. The synthetic-data ablation further shows that synthetic examples affect model performance; detailed results and discussion are provided in Appendix A.21.

Future work will focus on incorporating broader contextual information by expanding the dataset with additional data and contextual features, such as surrounding sentences and section information. This is expected to improve classification accuracy and provide a more holistic understanding of citance usage. Finally, we aim to explore applied use cases of this framework, as outlined in the appendix

(A.15).

Citance	By branching, the work of ResNeXt work of [89] can achieve much higher accuracy than the ResNet [25] at the same computational budget.
Citation Marks	[89], [25]
Predictions [89]	[Semantics: Artifact*/Methodology, Intent: Generic*/Comparison, Polarity: Supporting*/Refuting]
Predictions [25]	[Semantics: Artifact*/Methodology, Intent: Generic*/Comparison, Polarity: Refuting]

Figure 3: Challenging example leading to incorrect prediction.

Citance	MAGMA [5] provides powerful intelligently scheduled BLAS and LAPACK algorithms, but due to the dependency on external libraries is difficult to install, configure, and tune, and does not yet provide unified or consistent capability across its CUDA, OpenCL, and Intel MIC implementations.
Citation Mark	[5]
Prediction	[Semantics: Artifact, Intent: Generic, Polarity: Refuting]

Figure 4: Example of correct polarity interpretation.

Citance	Kourtellis et al. [24] extended [13] to fully dynamic and improved in both time and space together with a scalable distributed implementation.
Citation Mark	[24], [13]
Prediction [24], [13]	[Semantics: Methodology , Intent: Generic , Polarity: Neutral]

Figure 5: Successful prediction over implicit citation roles.

Citance	To measure HEL-specific antibodies, maxisorp plates (Nunc) were coated with 60 ml of recombinant HEL (10 mg/ml) diluted in NPP buffer (adapted from [21]).
Citation Mark	[21]
Prediction	[Semantics: Methodology, Intent: Reuse, Polarity: Neutral]

Figure 6: Successful prediction with subtle lexical cues.

Limitations

A notable limitation of our study is the omission of the context window around citance sentences during the dataset annotation and the training of our language models. We did not consider the text preceding and following the citance sentences, which

might limit the model’s ability to fully grasp and interpret citances within their broader textual environment. As citances often derive significant meaning from the surrounding text, which includes arguments, discussions, and conclusions that frame the cited work (MacRoberts and MacRoberts, 1984), this oversight could undermine the accuracy of our citance analysis and affect the model’s ability to identify subtle relationships within the text. Additionally, our approach did not utilize the specific sections of the papers from which citances were extracted. Recognizing whether a citance is from a section on methodology or related work, for example, could provide valuable insights into its intent, polarity, and semantics.

Furthermore, our dataset is imbalanced, reflecting real-world citance distributions, which may influence model behavior, particularly for minority classes. Another limitation is the size of our research team, which constrained our ability to assemble a larger corpus; although our current dataset of 1,165 examples yields strong performance, a more extensive dataset would likely improve overall results and reduce class-specific variance. Finally, due to limited computing resources, we were unable to fine-tune the full Llama 3.1 model and instead relied on a smaller 4-bit quantized version, and we could not conduct a comprehensive hyperparameter search to identify the optimal configuration for each model.

Finally, a limitation is the relatively small size of the test split, comprising only 244 citances (approximately 20.94% of the total 1,165). This limited sample size may reduce the statistical robustness of the evaluation and affect the reliability and generalizability of the results.

Ethics Statement

This research was conducted with strict adherence to ethical principles. We ensured that all data utilized were sourced from reputable databases and were handled with confidentiality and integrity. No personal or sensitive data were involved in this study. We also carefully considered the potential implications of our findings and methodologies to avoid unintended harm or bias. Our computational methods, including the use of algorithms and machine learning models, were designed and implemented to promote transparency and fairness. We acknowledge the importance of ethical considerations in academic research and strive to uphold

these values in our work.

Acknowledgements

This research was funded by the European Union’s Horizon Europe research and innovation programme under the AIXPERT project (Grant Agreement No. 101214389), which aims to develop an agentic, multi-layered, GenAI-powered framework for creating explainable, accountable, and transparent AI systems.

References

- Awais Athar. 2011. [Sentiment analysis of citations using sentence structure-based features](#). In *Proceedings of the ACL 2011 Student Session*, pages 81–87, Portland, OR, USA. Association for Computational Linguistics.
- Awais Athar and Simone Teufel. 2012. [Context-enhanced citation sentiment detection](#). In *Proceedings of the 2012 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pages 597–601, Montréal, Canada. Association for Computational Linguistics.
- Iz Beltagy, Kyle Lo, and Arman Cohan. 2019. [SciBERT: A pretrained language model for scientific text](#). In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, pages 3615–3620, Hong Kong, China. Association for Computational Linguistics.
- Steven Bird, Robert Dale, Bonnie Dorr, Bryan Gibson, Mark Joseph, Min-Yen Kan, Dongwon Lee, Brett Powley, Dragomir Radev, and Yee Fan Tan. 2008. [The ACL anthology reference corpus: A reference dataset for bibliographic research in computational linguistics](#). In *Proceedings of the Sixth International Conference on Language Resources and Evaluation (LREC’08)*, Marrakech, Morocco. European Language Resources Association (ELRA).
- Indra Budi and Yaniasih Yaniasih. 2023. [Understanding the meanings of citations using sentiment, role, and citation function classifications](#). *Scientometrics*, 128(1):735–759.
- Hyung Won Chung, Le Hou, Shayne Longpre, Barret Zoph, Yi Tay, William Fedus, Yunxuan Li, Xuezhi Wang, Mostafa Dehghani, Siddhartha Brahma, Albert Webson, Shixiang Shane Gu, Zhuyun Dai, Mirac Suzgun, Xinyun Chen, Aakanksha Chowdhery, Alex Castro-Ros, Marie Pellat, Kevin Robinson, Dasha Valter, Sharan Narang, Gaurav Mishra, Adams Yu, Vincent Zhao, Yanping Huang, Andrew Dai, Hongkun Yu, Slav Petrov, Ed H. Chi, Jeff Dean, Jacob Devlin, Adam Roberts, Denny Zhou, Quoc V. Le,

- and Jason Wei. 2024. [Scaling instruction-finetuned language models](#). *Journal of Machine Learning Research*, 25(70):1–53.
- Paolo Ciccarese, Elizabeth Wu, Gwendolyn T. Wong, Marco Ocana, June Kinoshita, Alan Ruttenberg, and Tim Clark. 2008. [The SWAN biomedical discourse ontology](#). *Journal of Biomedical Informatics*, 41(5):739–751.
- Arman Cohan, Waleed Ammar, Madeleine van Zuylen, and Field Cady. 2019. [Structural scaffolds for citation intent classification in scientific publications](#). In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*, pages 3586–3596, Minneapolis, Minnesota. Association for Computational Linguistics.
- Tim Dettmers, Artidoro Pagnoni, Ari Holtzman, and Luke Zettlemoyer. 2023. [QLoRA: Efficient fine-tuning of quantized LLMs](#). In *Advances in Neural Information Processing Systems*, volume 36, pages 10088–10115. Curran Associates, Inc.
- Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. 2019. [BERT: Pre-training of deep bidirectional transformers for language understanding](#). In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*, pages 4171–4186, Minneapolis, Minnesota. Association for Computational Linguistics.
- Aaron Grattafiori et al. 2024. [The Llama 3 herd of models](#). *Computing Research Repository*, arXiv:2407.21783. Version 3.
- Walid Hariri. 2024. [Sentiment analysis of citations in scientific articles using ChatGPT: Identifying potential biases and conflicts of interest](#). *Computing Research Repository*, arXiv:2404.01800. Version 3.
- Imran Ihsan and Muhammad Abdul Qadir. 2019. [CCRO: Citation’s context & reasons ontology](#). *IEEE Access*, 7:30423–30436.
- Imran Ihsan, Hameedur Rahman, Asadullah Shaikh, Adel Sulaiman, Khairan Rajab, and Adel Rajab. 2023. [Improving in-text citation reason extraction and classification using supervised machine learning techniques](#). *Computer Speech & Language*, 82:101526.
- Barbara Leitner, Joeri Tjink, Friederike Elisabeth Kohrs, Alexandra Bannach-Brown, Sven Arend Ulpts, Fakhri Momeni, Eleni Adamidi, Thomas Klebel, Eva Kormann, Adrian Marangoni, Jesper W. Schneider, Allyson L. Lister, Elli Papadopoulou, Haris Papageorgiou, Petros Stavropoulos, Stefania Amodeo, Thanasis Vergoulis, Susanna-Assunta Sansone, and Tony Ross-Hellauer. 2025. [Pilot implementation reflection report including assessment of efficacy and recommendations for future developments](#). Deliverable D4.3, TIER2 Project.
- Michael H. MacRoberts and Barbara R. MacRoberts. 1984. [The negational reference: or the art of dissembling](#). *Social Studies of Science*, 14(1):91–94.
- Quinn McNemar. 1947. [Note on the sampling error of the difference between correlated proportions or percentages](#). *Psychometrika*, 12(2):153–157.
- Silvio Peroni and David M. Shotton. 2012. [FaBiO and CiTO: Ontologies for describing bibliographic resources and citations](#). *Journal of Web Semantics*, 17:33–43.
- Marco Antonio Valenzuela-Escárcega, Vu A. Ha, and Oren Etzioni. 2015. [Identifying meaningful citations](#). In *Scholarly Big Data: AI Perspectives, Challenges, and Ideas*, pages 21–26, Austin, Texas, USA. AAAI Press.
- He Zhao, Zhunchen Luo, Chong Feng, Anqing Zheng, and Xiaopeng Liu. 2019. [A context-based framework for modeling the role and function of on-line resource citations in scientific literature](#). In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, pages 5206–5215, Hong Kong, China. Association for Computational Linguistics.
- Xiaodan Zhu, Peter D. Turney, Daniel Lemire, and André Vellino. 2015. [Measuring academic influence: Not all citations are equal](#). *Journal of the Association for Information Science and Technology*, 66(2):408–427.

A Appendix

A.1 Llama 3.1 / GPT-4o Mini System Prompt

Instructions: You are a citance analysis annotator. Your task is to analyze sentences from scientific publications containing the token <CIT>, which is the citance mark that refers to a specific cited work. Focus only on the part of the sentence directly associated with <CIT>. Ignore other citance marks unless they directly affect the interpretation of <CIT>. Do not focus on keywords as they may provide bias on the category, but focus on why the authors cites <CIT>, how the author feels towards <CIT> and how the author utilizes the work <CIT>. If there are many citance marks, ignore the meaning towards the other marks and focus only on the words that refer to <CIT>.

Classification Categories: For each sentence, select one option from each of the following categories based on the portion related to <CIT>: 1. Semantics: What is being cited? Claim: Specific statements, theories, or ideas. This is a very high overview of <CIT> work or a theoretic concept. Methodology: Procedures, techniques, or methods used. This is a high level overview and no

specific or distinct mentions to <CIT> methodology. Artifact: Tools, datasets, software, or other resources. This is a more detailed level, and there are specific mentions to <CIT> methodology with explicit detail. Results: Findings, data analyses, or conclusions from <CIT>. Choose the category that best represents the primary focus of the citance. 2. Intent: Why is the author citing it? Reuse: Direct use of elements from the cited work. Comparison: Highlighting similarities or differences between the author's current work and <CIT>. Extension: Building upon and enhancing <CIT> work. Generic: Providing general information from <CIT>. Select the option that most closely reflects the relationship between the author's work and the cited work. 3. Polarity: What is the author's sentiment? Supporting: Positive agreement or endorsement towards <CIT>. Can also be expressed through technical acknowledgment (state of the art, groundbreaking etc.). Neutral: No clear stance from the author towards <CIT>. Refuting: Disagreement or challenge towards <CIT>. Can also be expressed through technical critique.

A.2 Flan-T5 Prompt

You are analyzing sentences from scientific publications containing the token <CIT>, a citance mark referring to a cited work. Focus only on the part of the sentence directly associated with <CIT>. Ignore other citance marks unless they affect <CIT>. Avoid focusing on keywords; instead, focus on why the author cites <CIT>, how they feel towards <CIT>, and how they use <CIT>.

Classification Categories: 1. Semantics: What is being cited? Choose one: Claim (statements or theories), Methodology (general procedures), Artifact (detailed tools or methods), Results (findings). 2. Intent: Why is it cited? Choose one: Extension (expand or build upon), Reuse (direct usage), Comparison (author compares between his and <CIT> work), Generic (general info). 3. Polarity: What is the author's sentiment? Choose one: Supporting (can also be expressed through technical acknowledgment), Neutral, Refuting (Can also be expressed through technical critique).

A.3 Synthetic Data Creation System Prompt for GPT-4o

During the Synthetic dataset creation phase, we utilized ChatGPT with GPT-4o in order to further enrich our dataset. The following prompt was employed:

Your task is to generate in-text citance mentions (citances) for scientific publications. Each citance should fit within a specific classification based on the categories below: 1) Intent. Use one of the following answers: Reuse: citance involves directly using ideas or research findings or an artifact from the cited paper, treating them as if they are already true or established. This is the case if the author themselves are reusing the work of the citance as it is in the current research or will use it to experiment. Look for keywords: [Use, Experiment] or any synonyms of these. Comparison: citance is used to examine both the author's own work and the cited paper's work, observing how they are alike or different. Identify if the citance is compared with the author's work or with another citance. This is the case if the author themselves are comparing their work with the citance. Look for keywords: [Better, Worst, In contrast] or any synonyms of these words. Extension: citance is used to build upon and advance the idea and work of the cited paper. This is when the author cites a paper and then adds their own ideas to it as well, not just reusing the cited one. This is the case only if the author themselves are extending a work, not if the author mentions that another author extended a different work. Look for keywords: [Improve, Update, Enrich, Expand] or any synonyms of these words. Generic: citance is used to include basic info or general stuff from the cited paper that helps readers understand the author's work better. If the answer of question 1) is this, then the answer of question 2) can only be Neutral. 2) Polarity. Use one of the following answers: Supporting: citance is used as the author agrees with the findings, methods, or conclusions of the cited paper, and suggests that it aligns with the author's own perspective or contributes positively to their research. The positive sentiment could also be expressed through the technical acknowledgment of the cited work. Neutral: The author uses the citance to provide knowledge, show results, or to offer background without expressing a specific stance. It is used to establish context, acknowledge foundational research, or demonstrate the progression of ideas in the field, without indicating agreement or disagreement. Refuting: citance is used as the author challenges, refutes, or offers alternative perspectives to its work. Author uses it to indicate disagreement or divergence from the referenced paper, suggesting that the author's own research or viewpoint differs significantly or offers contrasting interpretations or solutions. The author

may still use a citance's work in their own work in order to refute it. The negative sentiment could also be expressed through the technical disregard of the cited work. 3) Semantics. Use one of the following answers: Claim: citance is used to back up, argue against or reference a claim by the cited paper. Look for keywords: [argue, back up, claim, address] or synonym of these words. Results: citance is used to show the results of the cited paper. Look for keywords: [proved, proven to, shown, indicate] or synonym of these words. Methodology: citance is used to show a method in the cited paper. If the sentence includes data, dataset, experiment on tools or software keywords are used, the answer is Artifact. Look for keywords: [used, leveraged, method] or synonym of these words. Artifact: citance is used to show that the citance refers to a tool from the cited paper. This could be data, datasets, lab equipment, surveys, medicine or software etc. used in the cited paper. Based on the specified classification below, generate a new citance that aligns with the given Polarity, Intent, and Semantics. The generated citances should always have a citance mark using the <CIT> placeholder, it should fit the specific classification indicated. The placeholder can be anywhere in the sentence, but logically it is where the synthetic sentence would require a citance The generated citance list should be numbered. Example: The economic models developed by Clark et al. suggest stable growth patterns, but the data from our latest study challenges this assertion, indicating increased volatility in market behaviors not captured by their original datasets. You should replace Clark et al. with <CIT>. Do not change the way you generate sentences. Write them as if you had Clark et al. in the first place then replace it, so the sentences are more realistic. Additionally, try to create realistic sentences. Do not use words such as "geographical" or "economical dataset" just to include the field of science. Make it more human and more natural. Number of citances to generate here: 15 Write the results in an excel (file ending with .xlsx). Include the following columns: id, paper_id, field_of_study, citance, citance_mark (the citance mark of the sentence), polarity, intent, semantics, manually (default value: null), synthetic (default value: Y) Classification: Intent: a) Extension b) Comparison c) Reuse Polarity: a) Refuting b) Supporting Semantics: a) Artifact b) Results Possible Field of Science: a) Health Sciences and Bio-informatics b) Computer Science c) Social Sciences and Humanities Generated citance

List: 1.

A.4 GPT-5 mini/GPT-4o mini - Few Shot Experiment System Prompt

Few Shot Examples

Citance Text: We compare our method against Make3D trained on the same datasets, as well as the published results of other current methods <CIT>. Classification: Semantics: Results Intent: Reuse Polarity: Neutral

Citance Text: <CIT> does not take into account different dataset inputs. Classification: Semantics: Methodology Intent: Generic Polarity: Refuting

Citance Text: Extension of <CIT> work, where we compare the use of partitioned versus global scheduling for MCS is the main contribution of our paper. Classification: Semantics: Methodology Intent: Extension Polarity: Supporting

Citance Text: Figure 5 shows the curves of the various models, and Comparison on VIPeR dataset Using the VIPeR dataset, we compared our method with such state-of-the-art methods as MtMCMML <CIT>, LMLF [13], SDALF [5], eBiCov [31], eSDC [32], PRDC [9], aPRDC [33], PCCA [14], KISSME [34], LF [35] and SalMatch [36]. Classification: Semantics: Artifact Intent: Comparison Polarity: Supporting

Citance Text: 'See also <CIT>, although I disagree with Huneman's reading of section 80 of the Critique of Judgment, insofar as it places Kant in opposition to the science of form. Classification: Semantics: Claim Intent: Generic Polarity: Refuting

A.5 GPT-5 mini / GPT-4o mini - Few-Shot Experiment Results

In addition to the prompt described in A.1, we evaluated a few-shot prompting approach using GPT-4o mini, as detailed in A.4. The prompt follows the same structure as in the aforementioned section, augmented with four training examples accompanied by explanations and five examples without explanations. To further investigate the behavior of reasoning-oriented models, we also evaluated GPT-5-mini using the same prompt. The performance of GPT-4o-mini and GPT-5-mini is shown in Table 4, where we also include Llama 3.1-FT for ease of comparison.

While GPT-4o-mini underperforms compared to Llama 3.1-FT, GPT-5-mini achieves comparable performance. Llama 3.1-FT outperforms

GPT-5-mini on the semantic classification task, arguably the most challenging dimension, while performance on intent and polarity is broadly similar. GPT-5-mini shows a slight advantage in intent classification, whereas Llama 3.1-FT performs better on polarity. SciBERT-MT surpasses GPT-5-mini in semantic and intent classification performance, while GPT-5-mini attains higher polarity accuracy. Although comparing a fine-tuned non-reasoning model with a reasoning model is not methodologically ideal, we include this experiment to provide additional comparative insight.

Citance analysis is typically conducted at large scale, often involving hundreds of thousands or millions of citances. Based on our GPT-5-mini measurements, the system processes approximately 8.13 citances per minute at a cost of \$0.0013 per citance. In contrast, Llama 3.1-FT required 5 minutes to process all 244 citances in the test set, corresponding to approximately 48.8 citances per minute, nearly six times faster than GPT-5-mini. Additionally, using SciBERT-MT, would require much less time, as inference for 244 citances took under 1 minute, corresponding to approximately 244 citances per minute. Moreover, running Llama 3.1-FT or SciBERT-MT incurs only electricity costs, as they operate locally and do not require API usage.

Semantic Scholar currently indexes over 200 million papers. Assuming an average of 30 references per paper, this corresponds to roughly 6 billion citances. Extrapolating our measurements to this scale, processing 6 billion citances with GPT-5-mini would require approximately 738 million minutes (approximately 1,400 years) of serial runtime and cost roughly \$7.9 million. While parallelization could reduce time, the overall monetary cost remains substantial. Using Llama 3.1-FT, processing the same volume would require approximately 123 million minutes (approximately 234 years) of serial runtime, but with much less monetary cost, as it can be run on local hardware without API fees. SciBERT-MT would require approximately 24.6 million minutes (approximately 47 years) of serial runtime, with similarly low monetary cost.

Consequently, although reasoning models are feasible for small to medium-scale studies, their computational requirements present significant challenges for very large-scale citance analysis. These constraints highlight the importance of our contribution for citance analysis: fine-tuning language models and developing a curated dataset that

enables scalable deployment.

A.6 Train-Test-Dev Split

To ensure fairness and eliminate bias in our model, we carefully maintained consistency across all possible combinations of polarity, intent, and semantics in different data subsets. Initially, we tried splitting papers into distinct sets, ensuring each paper appeared only once, but this approach disrupted the overall data distribution. Instead, we opted to use unique citances to divide the sets. To further enhance the model’s generalization ability, rare combinations and synthetic data were kept exclusively in the training set. For combinations that occurred infrequently (e.g., count=2), we split them by placing one in the training set and the other in the test set. The data was then split into 60.52% for training, 18.54% for development, and 20.94% for testing, ensuring balanced distribution and avoiding overfitting or bias toward any specific category. Lastly, we performed post-split validation using the RapidFuzz library⁵ to identify potential duplicates. Citances with a similarity score above 90% were flagged, and if found to be duplicates (based on different citance_mark identifiers), they were re-assigned to the set with fewer data points to maintain balance. As a final verification step, a manual inspection using fuzzy similarity scores was carried out to confirm the absence of any remaining duplicates below the defined threshold. Detailed steps are outlined in Appendix A.7.

A.7 Train Test Dev Split Rules

We added this section to provide the rules for the train test Dev split of the dataset in an easy to read manner. The goal of the split was to be around 60% for training, 20% for dev and 20% for test but due to the rules it could a little lower or higher for each category ($\pm 3\%$). The rules are:

1. The distribution of polarity, intent, and semantic categories is kept consistent across all sets, or as close as possible, while complying with the required set sizes.
2. Data points with the same ID were not placed in multiple sets. This was initially attempted using the paper_id, but due to insufficient frequency of certain combinations, it was not viable.

⁵<https://github.com/rapidfuzz/RapidFuzz>

Model	Semantic			Intent			Polarity		
	macro F1	weighted F1	accuracy	macro F1	weighted F1	accuracy	macro F1	weighted F1	accuracy
Llama 3.1-FT	80.66	80.90	81.15	74.55	77.73	77.87	63.89	72.92	77.46
SciBERT MT Class	77.39	78.25	78.28	79.41	81.42	81.97	58.59	69.04	75.41
GPT-5 mini - Few Shot	73.94	75.10	75.00	75.69	78.45	78.28	67.90	72.09	70.90
GPT-4o mini - Few Shot	66.72	67.58	67.21	63.72	66.39	66.39	63.69	66.72	65.16

Table 4: Performance comparison of Llama 3.1-FT, few shot GPT-4o mini and few shot GPT-5 mini across Semantic, Intent, and Polarity classification tasks.

3. All synthetic data are included in the training set.
4. Any rare cases (count = 1) for polarity, intent, and semantic combinations are always placed in the training set.
5. For less frequent cases (count = 2) of polarity, intent, and semantic combinations, one instance is placed in the training set and the other in the test set.
6. A post-split check using the RapidFuzz library⁶ is conducted to consider citances with a fuzzy matching score above 90% as potential duplicates across sets. If duplicates are found, they are assigned to the smaller set. In our case, citances with different citance marks are considered similar but not duplicates, so they are not removed.
7. A second post-split check ensures that no duplicate IDs appear across different sets, even after prior verification.

A.8 Model Training / Fine-Tuning Parameters

The hyperparameter configurations used for training and fine-tuning the proposed models are summarized in Table 5.

Model	Parameter	Value
SciBERT MT - Classifier	Hidden dimension	128
	Learning rate	2×10^{-5}
	Number of training epochs	4
	Train batch size	32
	Evaluation batch size	8
Llama 3.1 FT	Learning rate	9.53×10^{-5}
	Number of epochs	2
	LoRA rank (r)	4
	LoRA alpha	16
	LoRA dropout	0.149
	Batch size	2
Flan-T5	Gradient accumulation steps	2
	Learning rate	3×10^{-5}
	Number of training epochs	8
	Train batch size	16
	Evaluation batch size	16
	Warmup steps	10
	Weight decay	0.01

Table 5: Training and fine-tuning hyperparameters for the evaluated models.

⁶<https://github.com/rapidfuzz/RapidFuzz>

During inference, the Llama 3.1 FT model was evaluated using a temperature setting of 0 to ensure deterministic output generation.

A.9 FOS classification

This section presents the classification of fields of study into three major categories, based on their respective subfields. Table 7 provides details for Health Sciences and Bioinformatics, Table 8 for Computer Science and Table 6 provides details for Social Sciences and Humanities.

Social Sciences and Humanities

law
philosophy
experimental psychology
developmental & child psychology
archaeology
anthropology
art practice, history & theory
economics
languages & linguistics
communication & media studies
sociology
geography
general psychology & cognitive sciences
clinical psychology
religions & theology
education
social psychology
cultural studies
behavioral science & comparative psychology
applied ethics

Table 6: Fields of Study in Social Sciences and Humanities

Health Sciences and Bioinformatics
sport sciences
otorhinolaryngology
psychiatry
nuclear medicine & medical imaging
biomedical engineering
immunology
ophthalmology & optometry
virology
complementary & alternative medicine
environmental & occupational health
physiology
gastroenterology & hepatology
microscopy
general clinical medicine
obstetrics & reproductive medicine
endocrinology & metabolism
nutrition & dietetics
epidemiology
speech-language pathology & audiology
mycology & parasitology
bioinformatics
biochemistry & molecular biology
surgery
emergency & critical care medicine
toxicology
respiratory system
neurology & neurosurgery
arthritis & rheumatology
rehabilitation
substance abuse
biophysics
health policy & services
anesthesiology
microbiology
gerontology
legal & forensic medicine
dermatology & venereal diseases
pathology
anatomy & morphology
public health
general & internal medicine
orthopedics
oncology & carcinogenesis
pharmacology & pharmacy
genetics & heredity
tropical medicine
dentistry
geriatrics
allergy

Table 7: Fields of Study in Health Sciences and Bioinformatics

Computer Science
distributed computing
computer hardware & architecture
optoelectronics & photonics
energy
information systems
software engineering
networking & telecommunications
artificial intelligence & image processing
computation theory & mathematics
electrical & electronic engineering
medical informatics

Table 8: Fields of Study in Computer Science

A.10 Combination Information

The following are the combinations of Intent, Semantics, and Polarity categories for which synthetic data has been generated, and we have no real data.

- "Extension_Artifact_Refuting"
- "Extension_Artifact_Supporting"
- "Extension_Claim_Supporting"
- "Reuse_Artifact_Refuting"
- "Reuse_Results_Supporting"

Additionally, for the following categories we have no data:

- "Extension_Results_Refuting"
- "Extension_Claim_Refuting"
- "Reuse_Claim_Supporting"
- "Reuse_Claim_Refuting"
- "Reuse_Results_Refuting"

A.11 Statistics per Major Field of Study

In this section, statistics per major field of study and per category are presented in the following tables.

Row Labels	Count of id	% of total
Artifact	163	37.91%
Claim	58	13.49%
Methodology	188	43.72%
Results	21	4.88%

Table 9: Count and Percentage of Total citances for Computer Science per Semantics category.

Row Labels	Count of id	% of total
Artifact	130	28.51%
Claim	87	19.08%
Methodology	127	27.85%
Results	112	24.56%

Table 10: Count and Percentage of Total citances for Health Sciences and Bioinformatics per Semantics category.

Row Labels	Count of id	% of total
Artifact	53	19.00%
Claim	114	40.86%
Methodology	46	16.49%
Results	66	23.66%

Table 11: Count and Percentage of Total citances for Social Sciences and Humanities per Semantics category.

Row Labels	Count of id	% of total
Comparison	90	20.88%
Extension	44	10.21%
Generic	213	49.65%
Reuse	83	19.26%

Table 12: Count and Percentage of Total citances for Computer Science per Intent category.

Row Labels	Count of id	% of total
Comparison	93	20.39%
Extension	37	8.11%
Generic	221	48.46%
Reuse	105	23.03%

Table 13: Count and Percentage of Total citances for Health Sciences and Bioinformatics per Intent category.

Row Labels	Count of id	% of total
Comparison	42	15.05%
Extension	19	6.81%
Generic	169	60.57%
Reuse	49	17.56%

Table 14: Count and Percentage of Total citances for Social Sciences and Humanities per Intent category.

Row Labels	Count of id	% of total
Neutral	300	69.84%
Refuting	60	13.92%
Supporting	70	16.24%

Table 15: Count and Percentage of Total citances for Computer Science per Polarity category.

Row Labels	Count of id	% of total
Neutral	309	67.76%
Refuting	63	13.82%
Supporting	84	18.42%

Table 16: Count and Percentage of Total citances for Health Sciences and Bioinformatics per Polarity category.

Row Labels	Count of id	% of total
Neutral	217	77.78%
Refuting	33	11.83%
Supporting	29	10.39%

Table 17: Count and Percentage of Total citances for Social Sciences and Humanities per Polarity category.

A.12 Dataset Mapping Attempts

In an effort to facilitate comparison with existing citation-related datasets, we attempted to align our labeling ontology with those of SciRES, SciCite, and the polarity dataset from Athar et al. Below are the mappings we proposed:

SciRES Mappings

Semantics Mapping (left-side SciRES)

- Material → Artifact
- Method → Methodology
- Supplement → Claim

Intent Mapping (left-side SciRES)

- Use → Reuse
- Produce → Reuse
- Introduce → Generic
- Extend → Extension
- Compare → Comparison
- Other → Generic

SciCite Mappings (left-side SciCite)

- Background → Generic
- ResultComparison → Results or Comparison
- Method → Methodology or Artifact

Athar et al. Mappings (left-side Athar et al.)

- n → Negative
- o → Neutral
- p → Positive

A.13 Analysis Per Field

Across the three major domains of Computer Science (Table 18), Health Sciences and Bioinformatics (Table 19), and Social Sciences and Humanities (Table 20), performance of Llama 3.1-FT and SciBERT-MT exhibit some field-dependent patterns.

- **Computer Science** (Table 18): Performance in this domain shows a clear division of strengths between the two models. Llama 3.1-FT achieves better results in semantic classification, with higher Macro F1 (80.60% vs. 79.90%), Weighted F1 (88.69% vs. 87.76%), and accuracy (88.89% vs. 87.77%), indicating a stronger ability to capture citation semantics. Llama 3.1-FT also slightly outperforms SciBERT-MT in polarity classification in terms of Macro and Weighted F1, while both models reach the same accuracy. In contrast, SciBERT-MT performs better on intent classification, achieving higher Macro F1, Weighted F1, and accuracy. These results suggest that SciBERT-MT is particularly effective at modeling rhetorical intent in technical writing, while Llama 3.1-FT is stronger in semantic and polarity-related tasks.
- **Health Sciences and Bioinformatics** (Table 19): In this field, SciBERT-MT shows stronger performance in both semantic and intent classification. For semantics, it achieves higher Macro F1, Weighted F1, and accuracy, indicating more reliable identification of citation roles in biomedical texts. The advantage of SciBERT-MT is even more pronounced for intent classification, where it surpasses Llama 3.1-FT by a substantial margin across all metrics. In contrast, Llama 3.1-FT performs better on polarity classification, achieving higher

Macro F1, Weighted F1, and accuracy. This suggests that Llama is more sensitive to evaluative language and sentiment expression in biomedical writing, while SciBERT-MT remains better suited for structurally grounded tasks such as semantics and intent.

- **Social Sciences and Humanities** (Table 20): In this domain, Llama 3.1-FT clearly outperforms SciBERT-MT in both semantics and polarity. The difference is especially large for semantic classification, where Llama 3.1-FT exceeds SciBERT-MT by over 12% in Macro F1, along with substantially higher Weighted F1 and accuracy. For polarity, Llama 3.1-FT again achieves higher scores across all metrics, reflecting strong and well-balanced sentiment modeling. For intent classification, however, SciBERT-MT performs better, with higher Macro F1, Weighted F1, and accuracy. Overall, the strong performance of Llama 3.1-FT in this field likely reflects the expressive and argumentative nature of Social Sciences and Humanities writing, where evaluative language and discourse structure are more explicit and align well with the broad linguistic capabilities of large language models.

Category	Model	Macro F1 (%)	Weighted F1 (%)	Accuracy (%)
Semantics	Llama 3.1-FT	80.60	88.69	88.89
	SciBERT-MT	79.90	87.76	87.77
Intent	Llama 3.1-FT	68.93	73.79	74.44
	SciBERT-MT	70.14	75.08	76.66
Polarity	Llama 3.1-FT	53.39	52.41	63.33
	SciBERT-MT	49.63	52.37	63.33

Table 18: Comparison of Macro F1, Weighted F1, and Accuracy between Llama 3.1-FT and SciBERT-MT for each task in Computer Science. Each number is in % and the number of test samples (support) are 90.

Category	Model	Macro F1 (%)	Weighted F1 (%)	Accuracy (%)
Semantics	Llama 3.1-FT	77.06	76.55	77.38
	SciBERT-MT	80.71	80.37	80.95
Intent	Llama 3.1-FT	81.49	80.85	80.95
	SciBERT-MT	90.85	89.01	89.28
Polarity	Llama 3.1-FT	64.74	76.51	79.76
	SciBERT-MT	60.29	72.41	77.38

Table 19: Comparison of Macro F1, Weighted F1, and Accuracy between Llama 3.1-FT and SciBERT-MT for each task in Health Sciences and Bioinformatics. Each number is in % and the number of test samples (support) are 84.

Category	Model	Macro F1 (%)	Weighted F1 (%)	Accuracy (%)
Semantics	Llama 3.1-FT	74.35	77.08	75.71
	SciBERT-MT	61.77	63.36	62.85
Intent	Llama 3.1-FT	70.97	77.06	78.57
	SciBERT-MT	74.58	78.26	80.00
Polarity	Llama 3.1-FT	80.82	92.10	92.86
	SciBERT-MT	72.77	86.71	88.57

Table 20: Comparison of Macro F1, Weighted F1, and Accuracy between Llama 3.1-FT and SciBERT-MT for each task in Social Sciences and Humanities. Each number is in % and the number of test samples (support) are 70.

A.14 Analysis Per Class

The comparative evaluation of Llama 3.1-FT and SciBERT-MT across semantics, intent, and polarity classification tasks (see Tables 21, 22, 23) reveals distinct strengths and weaknesses for each model, providing guidance for their practical application in citation analysis. Overall, Llama 3.1-FT demonstrates stronger and more consistent performance in the Semantics and Polarity tasks, whereas SciBERT-MT maintains a clear and robust advantage in the Intent task.

- **Semantics Task** (Table 21): Llama 3.1-FT demonstrates stronger performance in most semantic categories, achieving higher F1-scores for **Artifact**, **Claim**, and **Results**. Its advantage is particularly clear for **Results** (85.37% vs. 78.05%) and **Claim** (71.29% vs. 66.07%), which require capturing more abstract and interpretive information from citation contexts. However, SciBERT-MT slightly outperforms Llama 3.1-FT on the **Methodology** class (78.26% vs. 77.78%), indicating a marginal advantage in identifying methodological references. Overall, Llama 3.1-FT attains higher accuracy (81.15%) and macro-average F1 (80.66%), reflecting stronger and more consistent performance across the semantic categories.
- **Intent Task** (Table 22): SciBERT-MT outperforms Llama 3.1-FT on most intent categories and across all aggregate metrics, achieving higher F1-scores for **Comparison**, **Extension**, and **Generic**. This highlights its strong reliability for detecting structured intent. Llama 3.1-FT slightly surpasses SciBERT-MT on the **Reuse** class in terms of F1-score (70.59% vs. 69.66%), but SciBERT-MT still maintains higher overall accuracy (81.97% vs. 77.87%) and macro-average F1 (79.41% vs. 74.55%).

Label	Precision (%)	Recall (%)	F1-Score (%)	Support
Llama 3.1-FT				
Artifact	87.65	88.75	88.20	80
Claim	83.72	62.07	71.29	58
Methodology	71.79	84.85	77.78	66
Results	83.33	87.50	85.37	40
Accuracy	81.15	81.15	81.15	244
Macro Avg	81.63	80.79	80.66	244
Weighted Avg	81.72	81.15	80.90	244
SciBERT-MT				
Artifact	89.47	85.00	87.18	80
Claim	68.52	63.79	66.07	58
Methodology	75.00	81.82	78.26	66
Results	76.19	80.00	78.05	40
Accuracy	78.28	78.28	78.28	244
Macro Avg	77.30	77.65	77.39	244
Weighted Avg	78.40	78.28	78.25	244

Table 21: Per-class performance and average metrics for Llama 3.1-FT and SciBERT-MT on the Semantics classification task in all fields. Scores are in % and best values between the models are bolded.

These results indicate that SciBERT-MT provides more robust and balanced performance for intent classification overall.

- **Polarity Task** (Table 23): Llama 3.1-FT exhibits conservative prediction behavior for the minority **Refuting** and **Supporting** classes, achieving very high precision for both (93.33% and 73.33%, respectively) but at the cost of lower recall. This indicates that while its predictions are reliable when made, it misses a substantial number of true instances, particularly for **Supporting** (20.37% recall). In contrast, SciBERT-MT achieves a more balanced performance for **Refuting**, with a higher recall (65.22%) and strong F1-score (71.43%), outperforming Llama 3.1-FT on this category. For **Supporting**, SciBERT-MT mirrors Llama’s conservative behavior and performs even worse in terms of recall (11.11%). Overall, Llama 3.1-FT achieves higher accuracy (77.46% vs. 75.41%) and weighted F1 (72.92% vs. 69.04%) due mainly to its strong performance on the dominant **Neutral** class.

A.15 Use Cases

Citance analysis enables value creation for four main stakeholder groups in the scholarly ecosystem: researchers, research-performing organizations, publishers, and funders. Each benefits in distinct but complementary ways from fine-grained citation context modeling.

- **Researchers:** Citance analysis provides researchers with powerful tools for knowledge

Label	Precision (%)	Recall (%)	F1-Score (%)	Support
Llama 3.1-FT				
Comparison	79.59	72.22	75.73	54
Extension	66.67	72.73	69.57	11
Generic	79.85	84.92	82.31	126
Reuse	73.47	67.92	70.59	53
Accuracy	77.87	77.87	77.87	244
Macro Avg	74.89	74.45	74.55	244
Weighted Avg	77.81	77.87	77.73	244
SciBERT-MT				
Comparison	89.58	79.63	84.31	54
Extension	75.00	81.82	78.26	11
Generic	79.05	92.86	85.40	126
Reuse	86.11	58.49	69.66	53
Accuracy	81.97	81.97	81.97	244
Macro Avg	82.44	78.20	79.41	244
Weighted Avg	82.73	81.97	81.42	244

Table 22: Per-class performance and average metrics for Llama 3.1-FT and SciBERT-MT on the Intent classification task in all fields. Scores are in % and best values between the models are bolded.

Label	Precision (%)	Recall (%)	F1-Score (%)	Support
Llama 3.1-FT				
Neutral	76.64	98.20	86.09	167
Refuting	93.33	60.87	73.68	23
Supporting	73.33	20.37	31.88	54
Accuracy	77.46	77.46	77.46	244
Macro Avg	81.10	59.81	63.89	244
Weighted Avg	77.48	77.46	72.92	244
SciBERT-MT				
Neutral	74.77	97.60	84.68	167
Refuting	78.95	65.22	71.43	23
Supporting	85.71	11.11	19.67	54
Accuracy	75.41	75.41	75.41	244
Macro Avg	79.81	57.98	58.59	244
Weighted Avg	77.59	75.41	69.04	244

Table 23: Per-class performance and average metrics for Llama 3.1-FT and SciBERT-MT on the Polarity classification task in all fields. Scores are in % and best values between the models are bolded.

discovery and research planning. By analyzing how and why prior work is cited, researchers can:

- Identify the most frequently reused datasets, methods, and artifacts within a specific research problem.
- Detect contradictions, debates, and unresolved disagreements in their scientific field.
- Discover gaps in the literature where evidence is missing, weak, or inconsistent.
- Support systematic reviews and meta-analyses by clustering citances according to their role.
- Improve reproducibility by tracing which works are reused or extended by later studies.

These capabilities support more informed research design, faster onboarding into new topics, and higher-quality scientific contributions.

• **Research-performing organizations (e.g.,**

universities, research institutes): Universities and research institutions are increasingly evaluated based on the impact, reuse, and societal relevance of their outputs. Citance analysis enables:

- Fine-grained research impact assessment beyond raw citation counts, capturing why an institution’s work is cited.
- Evidence of research reproducibility and reusability based on downstream reuse and positive community uptake.
- Improved institutional benchmarking and rankings through richer indicators of scientific influence.
- Stronger justification for public and private funding by demonstrating measurable impact.

This allows institutions to better showcase the real scientific value of their research portfolios.

• **Publishers and scientific venues.** Publishers can leverage citance-level analytics to strengthen the visibility, credibility, and strategic positioning of their journals and conferences. In particular, they can:

- Provide advanced analytics on how published articles are used, extended, or debated in the literature.
- Measure the influence and strengths of journals beyond traditional impact factors.
- Identify emerging research trends and controversial or rapidly developing topics.
- Enhance editorial decision-making and journal branding by demonstrating the real-world scientific influence of published work.

This increases both the perceived impact and the competitive standing of publication venues.

• **Funders and funding agencies.** Public and private funders invest billions of dollars into research but often lack fine-grained visibility into how funded results are actually used by the scientific community. Citance analysis enables:

- Direct measurement of the impact of the funded research over time.
- Quantitative and qualitative indicators of scientific impact, relevance, and community perception.
- Identification of highly influential projects emerging from funding programs.
- Evidence-based evaluation of funding effectiveness, helping answer the question: Are these funding investments producing meaningful scientific impact?

This supports more transparent, accountable, and strategic allocation of future research funding.

A.16 Inter-annotator agreement result analysis

For the Semantics dimension, a third annotator, who is pursuing a PhD in Artificial Intelligence with a focus on Natural Language Processing, was brought in to adjudicate ambiguous cases and determine whether the gold or the annotator labels were more accurate. This was necessary as several instances exhibited interpretive uncertainty between the gold and annotated labels. The confusion matrices in Tables 24, 25, and 26 summarize the disagreements between the gold standard and the annotator for Semantics, Intent, and Polarity, respectively.

Specifically, for 184 artifacts labeled as Artifact in the gold data, the annotator had labeled them as Methodology. The third annotator concluded that 77 instances (42%) were indeed more consistent with the Methodology category as the annotator had labeled them, while 107 instances (58%) were correctly labeled as Artifact in the gold data. Similarly, for 84 instances originally labeled as Result in the gold data but annotated differently, the third annotator determined that 70 cases (83.33%) were correct in the gold standard, 11 were better categorized as Claim, one as Methodology, and two as Artifact.

For the Polarity dimension, several mismatches were observed between the gold and annotator labels. In certain cases, gold instances labeled as Refuting were annotated as Supporting, and vice versa. Following detailed review by the third annotator, these inconsistencies were found to stem mainly from two sources: (1) confusion caused by multiple citation markers within a single sentence,

leading the annotator to associate the polarity with the wrong citation and (2) semantic misinterpretations of the sentence.

Finally, differences observed in the Intent category primarily arose from varying semantic interpretations between the annotator and the original authors regarding the intended meaning of the labels.

A.17 Absence of Within-Axis Multi-Label Cases

The annotation protocol allowed annotators to assign more than one label within the same axis, both to represent genuinely overlapping citation functions and to diagnose category overlap or weaknesses in the ontology. Nevertheless, no within-axis multi-label cases were observed. We attribute this outcome to the interaction between the sentence-level annotation unit and the ontology's operationally distinct category definitions.

First, the dataset consists of individual citation sentences, in which the relationship to the cited work was generally expressed through one dominant citation function. Citations can serve multiple purposes in a broader discourse context, but the instances in CitAn usually conveyed one primary relation clearly enough to support a single label within each axis. The absence of surrounding context may therefore have reduced the observable incidence of multiple independently expressed functions. Incorporating neighboring sentences or section-level context could reveal more multi-label cases and remains a direction for future work.

Second, the ontology was designed to minimize overlap among labels. Potential overlaps encountered during annotation generally involved implication rather than two functions being independently and explicitly expressed. The clearest example is Extension and Reuse: extending a cited work often presupposes some reuse, but the guidelines assign Extension when the cited work provides the basis for further development and reserve Reuse for direct application or adoption without a clear extension. This specificity rule allowed annotators to resolve such cases without assigning both labels. Thus, the absence of multi-label annotations should be understood as a property of the present sentence-level dataset and annotation guidelines, rather than evidence that scholarly citations can never serve multiple functions.

Gold / Annotator	Artifact	Claim	Methodology	Results
Artifact	235	4	173	9
Claim	44	65	107	30
Methodology	29	1	251	2
Results	8	20	20	166

Table 24: Confusion matrix for **Semantics**. Rows correspond to gold labels, and columns to annotator-assigned labels.

Gold / Annotator	Comparison	Extension	Generic	Reuse
Comparison	191	4	17	13
Extension	3	84	4	9
Generic	38	7	517	41
Reuse	15	2	30	190

Table 25: Confusion matrix for **Intent**. Rows correspond to gold labels, and columns to annotator-assigned labels.

A.18 Fine-tuning - Statistical Significance Testing

We evaluated whether fine-tuning significantly improved Llama 3.1 model performance using McNemar’s test (McNemar, 1947), a non-parametric test designed for comparing paired categorical predictions. The test was applied independently to each classification task (semantics, intent, and polarity) using paired predictions from the base and fine-tuned models evaluated on the same input instances. For each task, we constructed a 2×2 contingency table capturing the number of instances in which:

- both models were correct (n_{11}),
- both models were incorrect (n_{00}),
- only the base model was correct (n_{10}), and
- only the fine-tuned model was correct (n_{01}).

McNemar’s exact test focuses on the discordant pairs (n_{10} and n_{01}), which correspond to cases where the two models disagree. The null hypothesis (H_0) assumes that the base and fine-tuned models have equal probabilities of producing a correct prediction, such that any observed performance difference arises from random variation. The alternative hypothesis (H_1) posits that fine-tuning induces a systematic difference in predictive performance. Statistical significance was assessed using a two-sided McNemar’s exact test with a threshold of $\alpha = 0.05$.

Across all tasks (semantics, intent, and polarity), the fine-tuned model corrected more errors than it introduced ($n_{01} > n_{10}$), and all McNemar’s exact tests yielded p-values below 0.001, providing strong evidence that fine-tuning leads to a statistically significant improvement over the base model.

Gold / Annotator	Neutral	Refuting	Supporting
Neutral	688	51	87
Refuting	12	141	3
Supporting	58	8	117

Table 26: Confusion matrix for **Polarity**. Rows correspond to gold labels, and columns to annotator-assigned labels.

Task	n_{10}	n_{01}	p-value
Semantics	11	81	< 0.001
Intent	23	70	< 0.001
Polarity	20	52	< 0.001

Table 27: McNemar’s test results comparing base and fine-tuned models. n_{10} denotes cases where the base model was correct and the fine-tuned model was incorrect; n_{01} denotes cases where the fine-tuned model was correct and the base model was incorrect.

A.19 Llama 3.1-FT & SciBERT MT Classifier - Learning Curves

In Figure 7, we present the learning curve for Llama 3.1-FT, showing training and validation loss as a function of the number of training examples. Initially, we can see that the baseline loss is over 3 and rapidly decreases as the model learns from the training data, reaching a really low point after 200 training examples. However as seen in Figure 8, after approximately 400 training examples, the validation loss plateaus while the training loss continues to decrease, suggesting that the model has effectively learned from the available data and further training does not yield substantial generalization benefits. This indicates that Llama 3.1-FT achieves early convergence on this task.

In contrast, Figures 9 and 10 illustrate the learning dynamics of the SciBERT multi-task classifier. As training progresses, both training and validation loss steadily decrease, indicating consistent learning across epochs.

While the validation loss improves more gradually than in Llama 3.1-FT, it continues to decline throughout training, ultimately reaching its lowest value near the end of the fourth epoch. The zoomed view reveals that improvements persist even after 1,500 training examples, suggesting that SciBERT benefits from longer optimization and does not exhibit the early plateau observed with Llama 3.1-FT. The divergence between training and validation loss near the end of training suggests mild overfitting, however validation performance continues to improve, indicating that the model is still learning useful representations.

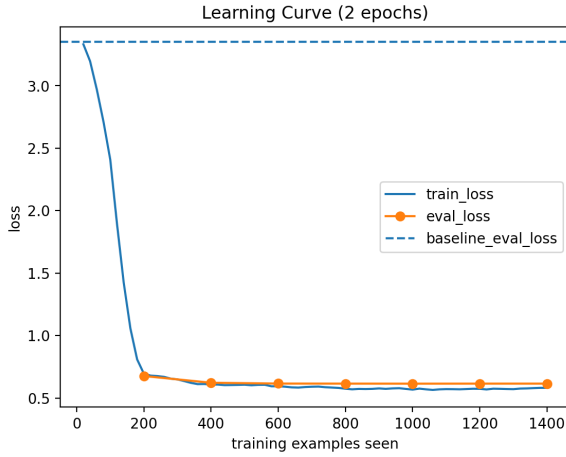


Figure 7: Learning curve for Llama 3.1-FT.

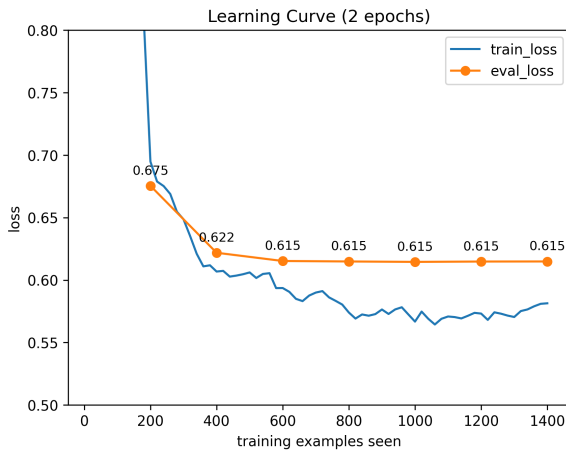


Figure 8: Zoomed learning curve highlighting validation loss improvements for Llama 3.1-FT.

To better capture task-level performance, Figure 11 presents the evolution of the custom evaluation metric (the sum of macro-averaged F1 scores across semantics, intent, and polarity tasks). The metric increases steadily throughout training, confirming that improvements in validation loss correspond to meaningful gains in predictive performance. Notably, performance gains continue beyond the midpoint of training before stabilizing, reinforcing the observation that SciBERT requires more training iterations to reach optimal generalization compared to Llama 3.1-FT.

Overall, these results highlight distinct learning behaviors: Llama 3.1-FT demonstrates rapid convergence and early saturation, whereas SciBERT exhibits slower but sustained improvement, benefiting from extended training to optimize multi-task performance.

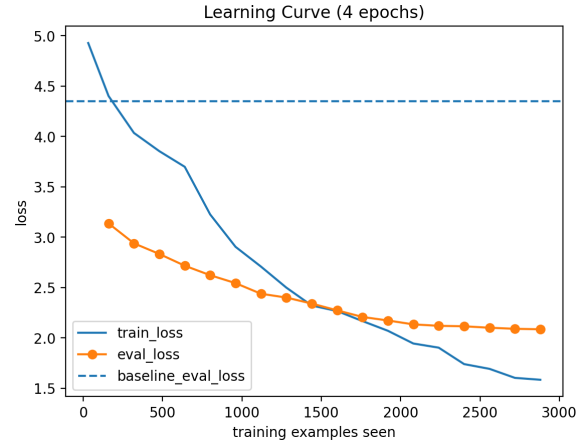


Figure 9: Full learning curve for the SciBERT multi-task classifier.

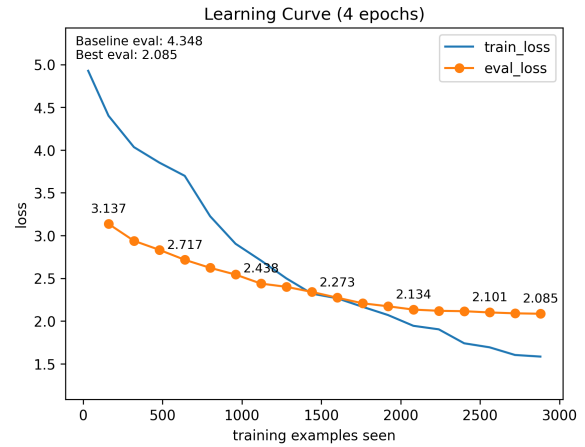


Figure 10: Zoomed learning curve highlighting validation loss improvements for SciBERT.

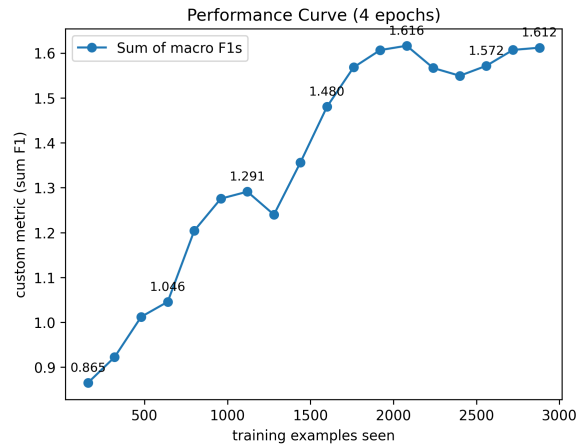


Figure 11: Performance curve showing the sum of macro F1 scores during SciBERT training.

A.20 Automation Bias Analysis

During corpus construction, Annotator A had access to the Llama 3.1 outputs and used them only

as a preliminary aid, adjusting labels where necessary. When resources became available, Annotator B subsequently annotated the resulting subset independently from scratch, without exposure to Llama outputs or Annotator A’s labels. The adjudicator was likewise blind to any LLM-generated suggestions. This design prevents the propagation of automation bias beyond the initial selection stage and provides an internal control for detecting potential influence. If exposure to Llama outputs had substantially biased Annotator A’s judgments, we would expect large systematic divergences between annotators. Instead, inter-annotator agreement was moderate for semantics ($\kappa = 0.44$) and substantial for intent and polarity ($\kappa = 0.76$ and $\kappa = 0.61$) - see Section 3.3. This indicates broadly consistent independent judgments rather than strong directional drift. Moreover, disagreements were distributed across categories rather than concentrated toward a single label, providing no evidence of systematic label skew attributable to LLM assistance. Together, the independent annotation procedure, blind adjudication, and agreement patterns suggest that the final annotations are unlikely to reflect automation bias.

A.21 Synthetic Data Impact on Model Performance

To evaluate the impact of synthetic data on model performance, we conducted an ablation study comparing Llama 3.1-FT and SciBERT-MT with and without synthetic examples in the training set. The results in Table 28 show that incorporating synthetic data improves Llama 3.1-FT on intent and polarity, whereas the model without synthetic data performs slightly better on semantics. Thus, the benefit of synthetic augmentation is task-dependent.

For SciBERT MT Class, the effect of synthetic data is even more pronounced, revealing a strong dependence on augmented training examples. Removing synthetic data leads to substantial degradation across all tasks, with the most severe decline observed in polarity classification, suggesting heightened sensitivity to data sparsity and class imbalance. More broadly, the findings indicate that the benefits of synthetic data are both model-dependent and task-sensitive: while Llama 3.1-FT shows its largest gains in intent and polarity detection, SciBERT improves across all tasks, implying that encoder-based models benefit more from increased training diversity. The consistent improve-

ments in macro F1 align with the goal of synthetic augmentation, namely strengthening performance on underrepresented classes and improving robustness. In particular, the notable increase in macro F1 for polarity classification suggests that minority classes are underrepresented in the training distribution, and as such their patterns are insufficiently learned without augmentation. This observation also reflects a broader characteristic of research writing, where authors tend to adopt neutral language, limiting the presence of strongly polarized expressions in the non synthetic data. Interestingly, Llama 3.1-FT without synthetic augmentation slightly outperforms the augmented variant on the semantic task, suggesting that the original dataset already provides sufficient semantic variability, thereby limiting the added value of synthetic examples for this task.

A limitation of this is that the hyperparameters were optimized using Optuna when synthetic data were included in the training set, thus, they may not be optimal for the non-augmented setting. Moreover, the original data split was preserved, meaning that removing synthetic examples reduced the effective training set size (580 of 705 instances). This preserves the evaluation and test splits and the original hyperparameters, but it also changes the amount of training data. Consequently, the comparison cannot fully separate the contribution of synthetic-data quality from that of additional training examples. We therefore interpret the SciBERT-MT ablation more cautiously: the observed performance difference likely reflects both reduced training-set size and the loss of synthetic examples that improve class balance and coverage of rare label combinations.

A.22 Statistical Analysis of Model Metrics

To assess the robustness and reliability of the evaluation results, we performed bootstrap resampling on the test set ($N=244$) with 5,000 iterations. This procedure estimates the variability of performance metrics that may arise from sampling differences in the test data. We report the mean performance along with 95% confidence intervals (CI) in Tables 29 and 30.

The fine-tuned Llama 3.1 model demonstrates strong performance on the *semantic classification* task, achieving a macro F1 score of 0.80 (95% CI: 0.75–0.85), indicating consistent identification of the cited content type. Performance on the *intent classification* task is slightly lower but remains

Model	Semantic			Intent			Polarity		
	macro F1	weighted F1	accuracy	macro F1	weighted F1	accuracy	macro F1	weighted F1	accuracy
Llama 3.1-FT	80.66	80.90	81.15	74.55	77.73	77.87	63.89	72.92	77.46
Llama 3.1-FT without Synthetic Data	81.15	81.93	81.97	66.72	74.01	74.59	53.32	67.18	74.18
SciBERT MT Class	77.39	78.25	78.28	79.41	81.42	81.97	58.59	69.04	75.41
SciBERT MT Class without Synthetic Data	63.90	65.44	65.16	50.77	68.19	70.90	31.13	57.39	68.85

Table 28: Performance comparison of macro and weighted F1 and accuracy of Llama 3.1-FT and SciBERT-MT with and without synthetic data during their training.

Task	Metric	Mean	Std	CI 2.5%	CI 97.5%
Semantic	Accuracy	0.811	0.025	0.762	0.861
Semantic	Macro F1	0.805	0.026	0.753	0.855
Semantic	Weighted F1	0.808	0.026	0.757	0.858
Intent	Accuracy	0.778	0.027	0.721	0.828
Intent	Macro F1	0.741	0.039	0.659	0.812
Intent	Weighted F1	0.777	0.027	0.721	0.828
Polarity	Accuracy	0.774	0.027	0.721	0.828
Polarity	Macro F1	0.637	0.041	0.540	0.713
Polarity	Weighted F1	0.728	0.034	0.662	0.794

Table 29: Bootstrap estimates (5,000 resamples) of evaluation metrics for Llama 3.1-FT. Confidence intervals represent the 95% percentile bootstrap interval.

robust, with a macro F1 score of 0.74 (95% CI: 0.66–0.81), suggesting reliable inference of citation purpose. The *polarity classification* task is the most challenging, with a macro F1 score of 0.64 (95% CI: 0.54–0.71). This lower score is expected due to the subtle and context-dependent nature of sentiment toward cited work, as well as potential class imbalance.

SciBERT-MT Class exhibits comparable robustness but a different performance profile. For *semantic classification*, SciBERT achieves a macro F1 score of 0.77 (95% CI: 0.72–0.82), indicating reliable recognition of citation content types, though slightly lower than Llama 3.1. On the *intent classification* task, SciBERT performs strongly with a macro F1 score of 0.79 (95% CI: 0.71–0.85), demonstrating stable inference of citation purpose. Similar to Llama 3.1, the *polarity classification* task remains the most difficult, with a macro F1 score of 0.58 (95% CI: 0.51–0.66), reflecting the nuanced and context-dependent nature of citation sentiment.

Across both models, the confidence intervals remain moderately narrow, indicating that performance is stable and not overly sensitive to the specific composition of the test set. These results provide evidence that the reported metrics are robust and reflect the models’ expected performance on comparable data distributions.

Task	Metric	Mean	Std	CI 2.5%	CI 97.5%
Semantic	Accuracy	0.782	0.027	0.730	0.832
Semantic	Macro F1	0.772	0.028	0.716	0.823
Semantic	Weighted F1	0.782	0.027	0.728	0.833
Intent	Accuracy	0.819	0.025	0.770	0.865
Intent	Macro F1	0.790	0.035	0.714	0.853
Intent	Weighted F1	0.814	0.026	0.762	0.863
Polarity	Accuracy	0.754	0.027	0.701	0.807
Polarity	Macro F1	0.584	0.038	0.508	0.656
Polarity	Weighted F1	0.690	0.036	0.619	0.759

Table 30: Bootstrap estimates (5,000 resamples) of evaluation metrics for SciBERT-MT Class. Confidence intervals represent the 95% percentile bootstrap interval.